Do Now · retrieval

Lesson 062 · Thinking Procedurally: Ordered Subtasks

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a precondition?

Q2

What is decomposition?

Q3

What is a dependency in a process?

Q4

An empty stack executes push(3), push(5), x=pop(), push(8), then push(x+peek()). peek() returns the top value without removing it. Trace the final stack from bottom to top.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Identify — Identify the separate components of a problem: validation, stock check, updating records and confirming a result.

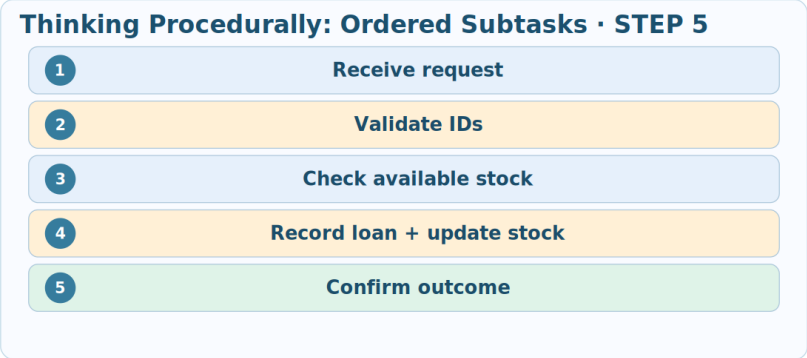
Core idea 2

Model — Determine necessary order from dependencies. Do not send success or deduct stock before validation.

Core idea 3

Evaluate — Decompose large operations into subprocedures, each with clear responsibility; update-related steps may need a single transaction to maintain consistency.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

procedural thinking — Planning ordered steps.

subprocedure — Smaller named piece of a solution.

dependency — A stage depends on another.

atomicity — All required updates succeed or none do.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

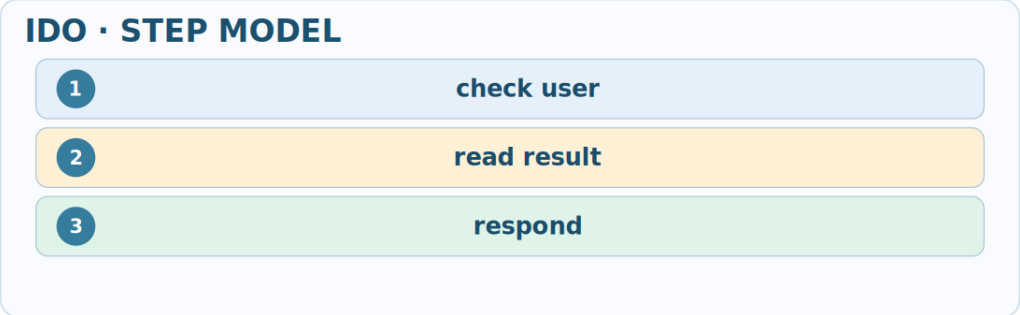
Problem

Put steps in order to process an exam result securely: check identity, find result, return data.

Follow the worked method

1. Explain the decision — If lookup occurs before checking authorisation, data can be exposed through insecure responses.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the decision

Payment and stock operations need failure handling; no dispatch before a confirmed order.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Order tasks: issue library loan, check stock, validate pupil, generate receipt. Explain first and last. [6]

Question 2

Write three helper procedure names for the library process, each with one responsibility. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A system sends confirmation before saving a booking, then crashes. Explain fault and propose safer steps. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A confirmation should be sent before a transaction succeeds, to make an application feel faster.

Use key vocabulary from memory

1. Explain procedural thinking in your own words.

2. Explain subprocedure in your own words.

3. Explain dependency in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.